THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 24	Let ages be $3x$, $5x$: $(3x+8):(5x+8)=2:3$ gives $x=8$
2	**(a)** 1350	One part $=$ ₹450
3	**(b)** 104	$13\times 360/45$
4	**(b)** 140	Combine to $a:b:c$ first
5	**(c)** 24	Remaining food $=(30-10)\times 120$ man-days
6	**(c)** 6	New water $=36\times 1/2=18$ L
7	**(c)** 18900	Wage per worker per day $=$ ₹150
8	**(d)** 12	Let ages be $2x$, $5x$: $(2x+9):(5x+9)=7:13$ gives $x=6$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$300\div 5=60$ per kg, and $60\times 8=480$; R is the property used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true but are about different situations; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take $x=3k$, $y=4k$ (1); ratio $=23k:8k=23:8$ (1).	1+1
12	Difference of ratio terms $=4$ parts (1); one part $=13$, numbers 91 and 39 (1).	1+1
13	Let ages be $7x$, $2x$ (1); solve $(7x+6):(2x+6)=8:3$ to get $x=6$, ages 42 and 12 years (1).	1+1
14	Take $x=5k$, $y=2k$ (1); ratio $=23k:4k=23:4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Remaining provisions $=(45-15)\times 240=7200$ man-days (1); men left $=180$ (1); days $=40$ (1).	1+1+1
16	Let milk $=3k$, water $=2k$ (1); $(2k+10)/(3k)=4/3$ gives $k=5$ (1); milk $=15$ L, water $=10$ L (1).	1+1+1
17	Students per degree $=12/40$ (1); total $=108$ (1); Maths $=24$ students (1).	1+1+1
	OR: Remaining provisions $=(40-10)\times 180=5400$ man-days (1); men left $=135$ (1); days $=40$ (1).	1+1+1
18	Let milk $=5k$, water $=3k$ (1); $(3k+12)/(5k)=6/5$ gives $k=4$ (1); milk $=20$ L, water $=12$ L (1).	1+1+1
	OR: Students per degree $=11/55$ (1); total $=72$ (1); Maths $=27$ students (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $7.5\times 20=150$ km (1). (ii) $260\div 20=13$ cm (1). (iii) $20\text{ km}=20{,}00{,}000$ cm, so $1:20{,}00{,}000$ (1). (iv) At 50 km/h: $150\div 50=3$ h (1); at 75 km/h: $150\div 75=2$ h (1).	1+1+1+2
20	Let ages be $3x$, $7x$ (1); $(3x+5):(7x+5)=1:2$ gives $x=5$ (2); ages 15 and 35 years (1); 5 years ago: 10 and 30, ratio $1:3$ (1).	1+2+1+1
	OR: Work left after 8 days $=(24-8)\times 30=480$ man-days (2); workers now $=20$ (1); more days $=24$ (1); total $=32$ days, delay $=8$ days (1).	2+1+1+1

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1y_1=x_2y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}}\times 100$ (percentage) instead of $\times 360^\circ$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to 360°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 162	Combine to $a:b:c$ first
2	**(d)** 27	Remaining food $=(24-6)\times 90$ man-days
3	**(b)** 10	New water $=45\times 2/3=30$ L
4	**(b)** 32400	Wage per worker per day $=₹180$
5	**(d)** 20	Let ages be $4x$, $7x$: $(4x+10):(7x+10)=2:3$ gives $x=5$
6	**(c)** 2100	One part $=₹300$
7	**(d)** 140	$21\times 360/54$
8	**(c)** 112	Combine to $a:b:c$ first
9	**(c)** A is true but R is false.	$4\times 5=20$ km, so A is true. R is false: 1 cm on the map represents 5 km.
10	**(d)** A is false but R is true.	For inverse proportion the product xy is constant, so A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Difference of ratio terms $=3$ parts (1); one part $=15$, numbers 75 and 30 (1).	1+1
12	Let ages be $8x$, $3x$ (1); solve $(8x+10):(3x+10)=2:1$ to get $x=5$, ages 40 and 15 years (1).	1+1
13	Take $x=7k$, $y=3k$ (1); ratio $=29k:9k=29:9$ (1).	1+1
14	Difference of ratio terms $=5$ parts (1); one part $=12$, numbers 108 and 48 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Remaining provisions $=(30-6)\times 300=7200$ man-days (1); men left $=200$ (1); days $=36$ (1).	1+1+1
16	Let milk $=4k$, water $=5k$ (1); $(5k+8)/(4k)=7/4$ gives $k=4$ (1); milk $=16$ L, water $=20$ L (1).	1+1+1
17	Students per degree $=8/24$ (1); total $=120$ (1); Maths $=20$ students (1).	1+1+1
	OR: Remaining provisions $=(50-20)\times 160=4800$ man-days (1); men left $=120$ (1); days $=40$ (1).	1+1+1
18	Let milk $=7k$, water $=3k$ (1); $(3k+15)/(7k)=8/7$ gives $k=3$ (1); milk $=21$ L, water $=9$ L (1).	1+1+1
	OR: Students per degree $=13/36$ (1); total $=130$ (1); Maths $=39$ students (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $15\times 24=360$ worker-days (1). (ii) $360\div 20=18$ days (1). (iii) $360\div 9=40$ days (1). (iv) Workers needed $=360\div 12=30$ (1); so $30-15=15$ more workers (1).	1+1+1+2
20	Let ages be $2x$, $5x$ (1); $(2x+6):(5x+6)=5:11$ gives $x=12$ (2); ages 24 and 60 years (1); 6 years ago: 18 and 54, ratio $1:3$ (1).	1+2+1+1
	OR: Work left after 5 days $=(20-5)\times 36=540$ man-days (2); workers now $=27$ (1); more days $=20$ (1); total $=25$ days, delay $=5$ days (1).	2+1+1+1

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1y_1=x_2y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}}\times 100$ (percentage) instead of $\times 360^\circ$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to 360°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 28800	Wage per worker per day $= ₹120$
2	**(a)** 20	Let ages be $5x$, $9x$: $(5x+8):(9x+8)=7:11$ gives $x=4$
3	**(d)** 800	One part $= ₹200$
4	**(d)** 120	$22 \times 360/66$
5	**(c)** 42	Combine to $a:b:c$ first
6	**(c)** 20	Remaining food $= (20-4)\times 200$ man-days
7	**(a)** 21	New water $= 60 \times 3/5 = 36$ L
8	**(c)** 22050	Wage per worker per day $= ₹175$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{90}{360} \times 100 = 25\%$; R is the basis of the calculation.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$60 \div 12 = 5$ per pencil and $5 \times 18 = 90$ so A is true; R is also true but unrelated to the pencil cost.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let ages be $9x$, $4x$ (1); solve $(9x+8):(4x+8)=11:6$ to get $x=4$, ages 36 and 16 years (1).	1+1
12	Take $x=5k$, $y=6k$ (1); ratio $=36k:17k=36:17$ (1).	1+1
13	Difference of ratio terms $=5$ parts (1); one part $=7$, numbers 56 and 21 (1).	1+1
14	Let ages be $5x$, $2x$ (1); solve $(5x+14):(2x+14)=7:4$ to get $x=7$, ages 35 and 14 years (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Remaining provisions $=(36-12)\times 210 = 5040$ man-days (1); men left $=180$ (1); days $=28$ (1).	1+1+1
16	Let milk $=2k$, water $=7k$ (1); $(7k+6)/(2k)=10/2$ gives $k=2$ (1); milk $=4$ L, water $=14$ L (1).	1+1+1
17	Students per degree $=15/50$ (1); total $=108$ (1); Maths $=27$ students (1).	1+1+1
	OR: Remaining provisions $=(28-7)\times 96 = 2016$ man-days (1); men left $=72$ (1); days $=28$ (1).	1+1+1
18	Let milk $=6k$, water $=5k$ (1); $(5k+20)/(6k)=9/6$ gives $k=5$ (1); milk $=30$ L, water $=25$ L (1).	1+1+1
	OR: Students per degree $=11/44$ (1); total $=90$ (1); Maths $=33$ students (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{96}{240}\times 360^\circ = 144^\circ$ (1). (ii) $\frac{60}{240}\times 100 = 25\%$ (1). (iii) $\frac{48}{240}\times 360^\circ = 72^\circ$ (1). (iv) Others: $\frac{36}{240}\times 100 = 15\%$ and 54° (1); $144+90+72+54=360$ (1).	1+1+1+2
20	Let ages be $3x$, $8x$ (1); $(3x+14):(8x+14)=1:2$ gives $x=7$ (2); ages 21 and 56 years (1); 14 years ago: 7 and 42 , ratio $1:6$ (1).	1+2+1+1
	OR: Work left after 7 days $=(21-7)\times 28 = 392$ man-days (2); workers now $=14$ (1); more days $=28$ (1); total $=35$ days, delay $=14$ days (1).	2+1+1+1

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